ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION WINTER SEMESTER, 2023-2024

DURATION: 2 HOURS FULL MARKS: 120

CSE 4107: Structured Programming I

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

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1. a) With the help of an example C code, identify Directives, Functions, Statements, Blocks, and String Literals. 7 (CO1) (PO1)

b) Show the output produced by the code in Code Snippet 1. 10 (CO1) (PO1)

```
1  #include <stdio.h>
2  int main()
3  {
4      int i, j, k;
5      i = 5; j = 4; k = 5;
6      printf("%d \n", i > j || ++j < k);
7      printf( "%d %d %d\n", i, j, k);
8
9      i = 7; j = 8; k = 9;
10     printf("%d \n", i = 0 && j++ < k);
11     printf("%d %d %d\n", i, j, k);
12
13     i = 1; j = 3; k = 4;
14     printf("%d \n", --i || ++j - k && ++k);
15     printf( "%d %d %d\n", i, j, k);
16
17     i = 1; j = 3; k = 4;
18     switch ((i + j) % j)
19     {
20         case 0: printf("Zero");
21         case 1: printf("One");
22         case 2: printf("Two");
23     }
24     return 0;
25 }
```

Code Snippet 1: A C program for Question 1.b

c) Write a C program to calculate the Lowest Common Multiple (LCM) of two integer numbers a and b. The LCM of two numbers a and b can be calculated as $\text{LCM}(a, b) = \frac{a \times b}{\text{GCD}(a, b)}$, where GCD denotes the greatest common divisor. A sample input and output for the program in shown in Table 1. 13 (CO1) (PO1)

Table 1: Sample input and output LCM for Question 1.c

Sample Input	Sample Output
Enter two numbers: 10, 15	LCM of 10 and 15 is: 30
Enter two numbers: 12, 16	LCM of 12 and 16 is: 48

2. a) Determine the values of all the variables for the following `scanf` function, for the input: `Test1-20.3-4.0e3`. Explain your answer. 7 (CO1) (PO1)

```
scanf("Test%d%d%f%f", &i, &j, &x, &y)
```

b) The following code in Code Snippet 2 was supposed to check whether the number `num` is prime. But it does not print the expected output for some specific values of `num`. 2 + 8 (CO1) (PO1)

```
1  #include<stdio.h>
2
3  int main()
4  {
5      int num = 17, d = 0;
6      scanf("%d", &num);
7      for ( int d = 1; d <=num; d++ )
8      {
9          if ( num % d)
10             break;
11     }
12     if ( d <= num )
13         printf("Not prime \n");
14     else
15         printf("prime \n");
16
17     return 0;
18 }
```

Code Snippet 2: An erroneous 'PrimeCheck' C program for Question 2.b

i. Find the output of the code in Code Snippet 2 for the input of `num = 14`.

ii. Fix the code in Code Snippet 2 with minimal changes so that it can check prime numbers greater than or equal to 2.

c) Write a C program that asks the user to enter the amount of taxable income M, and then display the total due tax T by using the tax information from Table 2 and following these steps: 13 (CO1) (PO1)

1. Identify which tax brackets the income M falls into.
2. Calculate tax for each bracket, using the base amounts and rates specified in Table 2.
3. Subtract the taxable amount from the income to find the residual for the next bracket.
4. Repeat until the entire income has been accounted for.
5. Sum all the calculated taxes to find the total tax due T.

A sample input and output with explanation is shown in Table 3.

Table 2: Income Tax rate for Question 2.c

Tax Bracket	Income Range	Amount of tax
B_1	Not over \$750	1% of income
B_2	\$751 – \$2,250	\$7.50 plus 2% of amount over \$750
B_3	\$2,251 – \$3,750	\$37.50 plus 3% of amount over \$2,250
B_4	\$3,751 – \$5,250	\$87.50 plus 5% of amount over \$3,750
B_5	Over \$5,250	\$147.50 plus 7% of amount over \$5,250

Table 3: Sample input and output due tax calculation with explanation for Question 2.c

Sample Input	Sample Output
`Enter Total Income: $3200`	`Total Due Tax: $111.000`
Explanation	
$3,200 falls within the range $2,251 - $3,750 So, tax for the amount ($3200 − $2,250) = $950 is $37.50 + $950 × 0.03 = $66.00 Residual amount is ($3200 − $950) = $2,250 which falls within the range $751 - $2,250 Next tax for the amount ($2,250 − $750) = $1500 is $7.50 + $1500 × 0.02 = $37.50 Next Residual amount is ($2,250 − $1500) = $750 which falls in the range Not over $750 Next tax for the amount $750 is $750 × 0.01 = $7.50 Total tax: $66.00 + $37.50 + $7.50 = $111.00	

3. a) What is the difference between `i++` and `++i` ? 3 (CO1) (PO1)

b) With appropriate example, explain the role of `break` statements in `switch` statements. 4 (CO1) (PO1)

c) Show the output produced by the code in Code Snippet 3. 10 (CO1) (PO1)

```
1  #include <stdio.h>
2  int main()
3  {
4      int x = 128;
5      while (x > 0)
6      {
7          x /= 2;
8          if (x== 4)
9          {
10             printf("Skip %d\n", x);
11             continue;
12         }
13         printf("%d\n", x);
14     }
15     return 0;
16 }
```

64, 32, 16, 8, 4, 2, 1

Code Snippet 3: A C program for Question 3.c

d) Write a C program that asks the user to enter a number n, then prints the square of all even numbers between 1 and n (inclusive). 13 (CO1) (PO1)

Table 4: Sample input and output squares for Question 3.d

Sample Input	Sample Output
`Enter the upper limit: 200`	4 16 36 64 100 144 196

4. a) Write short notes on the following topics with examples: 1×7 (CO1) (PO1)

i. Signed Integer
ii. Unsigned Integer
iii. Decimal Integer Constant
iv. Octal Integer Constant
v. Hexadecimal Integer Constant
vi. Integer Overflow
vii. Floating Constant

b) Rearrange the lines of code in Code Snippet 4 so that it prints the shape in Figure 1. 10 (CO1) (PO1)

```
1  #include<stdio.h>
2  int main()
3  {
4  {
5  return 0;
6  int n = 5;
7  for(int j = 0; j<=2*row; j++)
8  for(int row=0 ; row<n; row++)
9  for(int i = 0; i<n-row; i++)
10 printf("*");
11 printf("\n");
12 printf(" ");
13 }
14 }
```

Code Snippet 4: A shuffled C program for Question 4.b

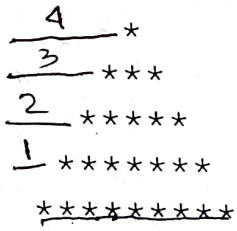


Figure 1: Pyramid shape for Question 4.b

c) Write a C program that asks the user to enter sixteen integer values (at most two digits) and then display the numbers in a 4 by 4 arrangement sequentially, followed by the sum of two diagonals. Follow the sample input output for better understanding. 13 (CO1) (PO1)

Sample Input:

```
Enter the 16 numbers:  1 20 13 4 5 6 7 8 9 10 11 12 13 24 25 16
```

Sample Output:

```
  1   20   13   4
  5   6    7    8
  9   10   11   12
 13   24   25   16
Diagonal Sums:  34 34
```